

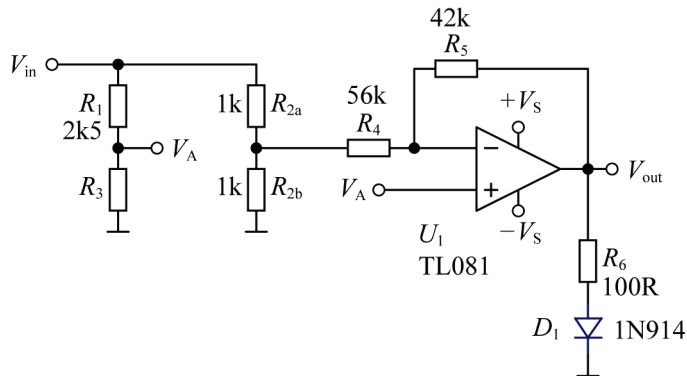
Project report

1. Circuit design considerations [2p]

Max. ½...1 page: General design considerations based on the specified devices and task description. Read carefully the **General notes**.

2. Circuit schematics [5p]

Max. ½...1 page: Uniquely annotate all components (R_1 , R_2 , etc.) and assign values (1k, etc.). Values of R and C can be given shorthand with their prefixes (e.g. $1.2\text{ k}\Omega = 1\text{k}2$, $100\text{ }\Omega = 100\text{R}$, $47\text{ nF} = 47\text{n}$) Type of special devices (sensors, op-amps, diodes, etc.) may be given only in the bill of materials. Build your circuit from the components we have learned (resistors, capacitors, diodes, Zeners, op-amps and BJTs). Select simple components first and only select special types if necessary (e.g. with high supply voltage, or large output current). Define ground and include input and output nodes as well as the power supplies. Wires of connected nodes can be replaced with unique node names (see V_A) if wiring is complicated. You may draw the circuit by hand, program, or using the editor of LTSpice: the visible image must be imported into the report.



3. Operation of the circuit [5p]

Max. 1 page: Details the operation of the circuit with loop/node analysis. Include the operation of the specified **input/output device as part of the circuit**. The circuits in the projects are large signal circuits with DC output that makes analysis simpler. Give the **transfer function** of the circuit as requested in the project. Design circuits as simple as possible to be able to entirely analyse them. For each project, you may find examples on the net, but in most cases without thorough analysis. If you select a component, look for its datasheet: e.g. you can simply give the opening voltage of a diode but with that estimation based on its datasheet for a given operating point. Considerations for selecting a specific component (e.g. a specific op-amp instead of a simple one because it has larger output current).

Explain briefly, how you generate **reference voltages and/or currents**.

Power supply considerations have a separate chapter.

4. Circuit transfer characteristics [3p]

Include here the plots of the circuit transfer characteristics, as requested in the task description. Mark the specified operating/discriminating points. If there are only a few such points e.g. in a switching application, calculate/estimate the characteristics within the ranges (e.g. a fan operating with only two speeds may operate with a small current (instead of zero) in its off state, and a slightly changing (instead of constant) current in its on state).

5. Bill of materials (BOM) [2p]

Max. 1 page: Give a tabular list of all components with their designators, values and quantities relevant for the circuit. Give also a short description, name the manufacturer and provide a link to the component datasheet (do **not** upload any datasheet with your report). A possible BOM is shown below.

Designator	Value	Description	Quantity	Manufacturer	Datasheet
R2a, R2b	1k	resistor	2	Vishay	link of source
U1	TL081	single op-amp	1	TI	link of source
...					

6. Power supply requirements [3p]

Max. ½...1 page: Summary of supply requirements based on the task specifications and circuit operation:

- Single, or dual supply, supply voltages, minimum (steady state) and maximum (on-state) currents.
- Calculation of power requirement of the circuit: at the input stage, at the op-amp stage and at the output drive stage. Summary of maximum power required per each supply.